

First, compared to the a-Si (conventional silicon series materials), the electron can easily pass through the IGZO circuit. The electron mobility is 20 to 50 times faster than the a-Si, resulting in the physical size of IGZO TFT being 1/7th of the a-Si TFT. This helps in achieving very high resolution.

Second is low power consumption. As we know the LCD runs at 60Hz in one frame. When the IGZO is turned on we only need a part of the frame and not the whole, resulting in a lot of power being saved.

Third, we achieve a LCD bezel. For instance, in a-Si TFT, the gate driver (COG) is at one side of the LCD, which cannot minimise the LCD side border. But for IGZO, monolithic gate drivers are used to build on both sides of the LCD, thus achieving a narrow frame.

Then there is the touch display. The touch and LCD signals are bound to interfere with each other, causing a problem. This is avoided in IGZO panels, as the touch detection can be activated when the IGZO TFT is in a rest mode. When the LCD is off, the touch is turned on and vice versa.

Reflective LCDs are eco friendly

The beauty of reflective LCD panels is their suitability for outdoor displays. Transmissive displays use a light source as the backlight, which, unfortunately, because of the ambient light, dulls the display. However, in reflective displays, the ambient light is utilised instead of a backlight. The features in a reflective IGZO are:

1. High visibility and full colour display
2. Low power consumption since there is zero backlight power
3. Video support since it runs at 60Hz
4. Easy on the eyes; usage of natural light can avoid strain to the eyes
5. Implementation of backlight

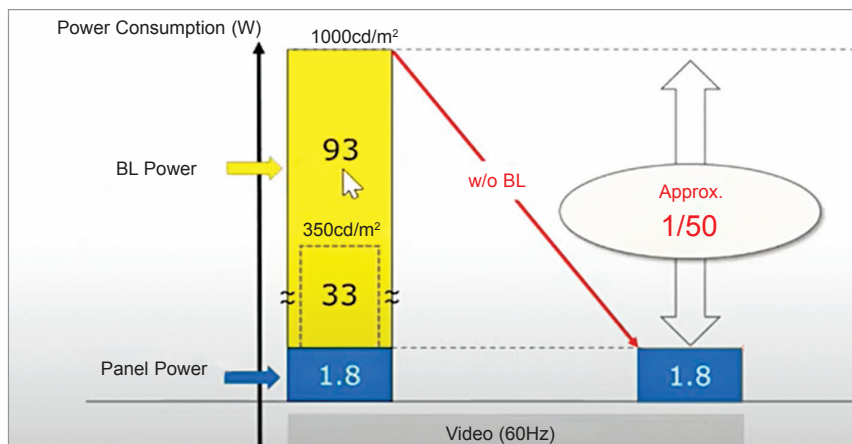


Fig. 6: Power consumption for transmissive and reflective displays

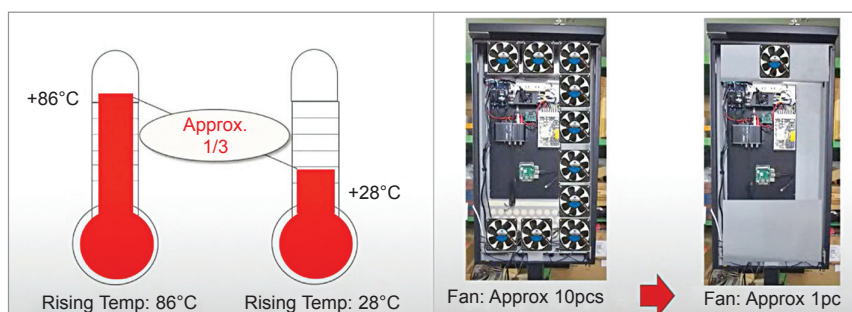


Fig. 7: Heat generated, and fans needed for transmissive and reflective displays

is optional; usage is promoted in a dark room

With no backlight being needed, we notice from the figures that around 95% of the power consumed is reduced. Similarly, the amount of heat generated is also reduced, leading to the fact that the number of fans needed to cool down the LCD drastically decreases. This makes the reflective LCDs much more reliable.

The remarkable fact is that, even an 80cm (31.5-inch) LCD can be run on a mobile battery (2500mAh) with a video being played for 24 hours without charging. Solar panels provide the freedom to move the system to different places. The solar panels along with the mobile batteries can help in signages located at bus stops, avoiding a lot of trouble. Not only bus stops, but this can also be implemented at petrol stations, vending machines, etc. There are various panel sizes available as sam-

ples and for mass production.

Regarding interfaces, there exist RGB and LVDS interfaces in the market, for many years. The RGB interface (for low-resolution panels) is mainly used for cars and motorcycles. The LVDS interface (for high resolution) is commonly used for industrial usage. Smartphones are using MIPI interface (low power consumption). EDP (power saving) is widely being used for laptop and monitor panels.

With such advantages that LCDs offer, design engineers have a tough job in choosing the right one. David suggests that it is best to match a design with different systems and interfaces before making the final selection. LCDs surely have the potential of being chosen as the best option for any application. **EFY**

The article is based on a tech talk by David Yang, Deputy General Manager, SHARP, Singapore at TechWorldCongress.com. Transcribed by Sharon Abhignya Katta.